

Challenge (Habanero)

- Q1.** If $2^x = 32$, what is the value of x ?
- (A) 3
(B) 4
(C) 5
(D) 6
(E) 7
- Q2.** A group of hikers is planning a trip to a remote area. They estimate that their fuel consumption will increase by a factor of 3^2 due to the rough terrain. If their initial fuel consumption was 4 gallons per hour, what will be their new fuel consumption?
- (A) 12 gallons/hour
(B) 16 gallons/hour
(C) 20 gallons/hour
(D) 24 gallons/hour
(E) 36 gallons/hour
- Q3.** The population of a certain animal species is growing exponentially according to the equation $P(t) = 2^t \cdot P_0$, where P_0 is the initial population. If the population doubles every 2 years, what is the value of t when the population reaches 4 times its initial value?
- (A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q4.** Solve for x : $x^2 \cdot 2^x = 64$
- (A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q5.** A park ranger estimates that the number of visitors to a park will increase by a factor of 5^2 over the next 5 years. If the current number of visitors is 100, what will be the estimated number of visitors in 5 years?
- (A) 500
(B) 1000
(C) 1200
(D) 1500
(E) 2500
- Q6.** If $3^x = 81$, what is the value of x ?
- (A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q7.** A group of friends plan to go on a road trip. They estimate that their fuel consumption will decrease by a factor of 2^2 if they drive at a constant speed of 60 mph. If their initial fuel consumption was 6 gallons per hour, what will be their new fuel consumption?
- (A) 1.5 gallons/hour
(B) 2 gallons/hour
(C) 3 gallons/hour
(D) 4 gallons/hour
(E) 6 gallons/hour
- Q8.** Solve for y : $y \cdot 4^y = 256$
- (A) 2
(B) 3
(C) 4
(D) 5
(E) 6

Open Ended Questions (FRQ)

- Q9.** A certain species of tree is growing in a forest. The number of trees is increasing exponentially according to the equation $N(t) = 3^t \cdot N_0$, where N_0 is the initial number of trees. If the number of trees doubles every 3 years, what is the value of t when the number of trees reaches 8 times its initial value? Show your work and explain your reasoning.
- Q10.** A traveler is planning a trip to a remote area. She estimates that her fuel consumption will increase by a factor of 2^x due to the rough terrain. If her initial fuel consumption was 8 gallons per hour and she wants to know her new fuel consumption when $x = 3$, what will be her new fuel consumption? Show your work and explain your reasoning.

Chef's Tips

- Emphasize the importance of understanding exponential growth and decay in real-world contexts.
- Encourage students to use mathematical models to solve problems and make predictions.
- Remind students to show their work and explain their reasoning when solving problems.